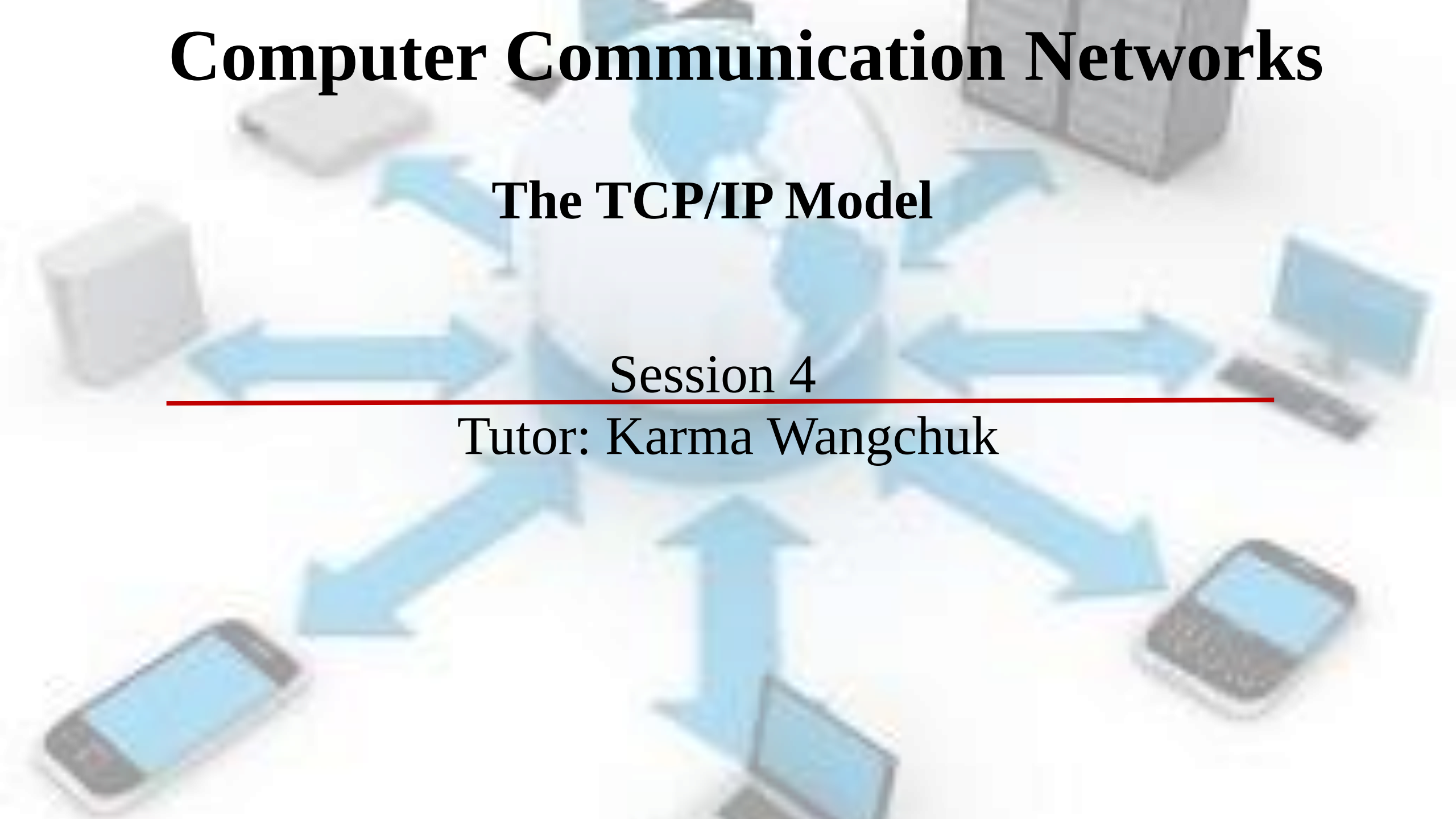
Computer Communication Networks

The TCP/IP Model

Session 4

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The TCP/IP Model

Learning Outcome:- At the end of the lesson, you are expected to be able to:

- Describe the TCP/IP Internet Model
- Differentiate between TCP/IP and OSI model
- Evaluate similarity between TCP/IP and OSI model



The TCP/IP Model: Introduction

- **Introduction: Brief history of Evolution of Internet**
  - ARPANet initially included only US government organizations and other companies were not allowed to access Internet but:
    - **1991:** Lifted the ban of commercial usage of the Internet.
    - **1994:** The first commercial web browser was released as a free product by Netscape Navigator followed by IE browser.
    - **1995:** The web content was developed and made accessible to the public.
    - **1996:** The search engine was developed and used.
    - **1997:** The push and pull technologies were launched, java applets, etc.



The TCP/IP Model: Introduction

Introduction:

- After Internet ban was lifted, hundreds of universities and government agencies were connected.
  - ✓ When **satellite** and **radio** networks were added later, the existing protocols had trouble interworking with them.
    - Therefore, TCP/IP Reference model was developed.
    - In order to create a network that survive any conditions.

- It was created by US DoD (Department of Defense)



The TCP/IP Model: Introduction

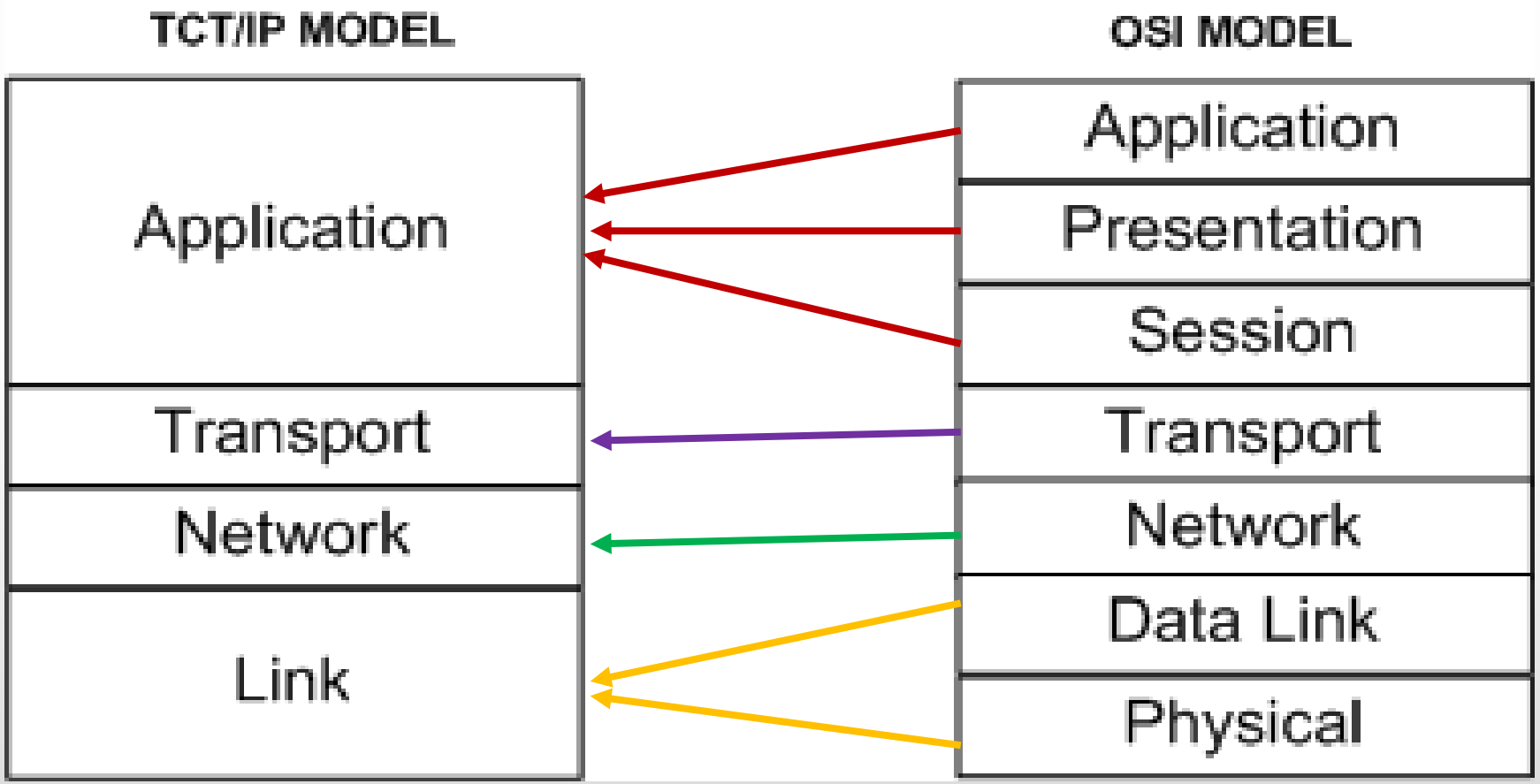
Introduction:

- The TCP/IP Model has 4 layers.
  1. Link Layer
  2. Network Layer
  3. Transport Layer
  4. Application layer
- It is also called Internet Model
- It is protocol specific model
- Some of the layers in TCP/IP model has same name as OSI model



The TCP/IP Model: Mapping 1

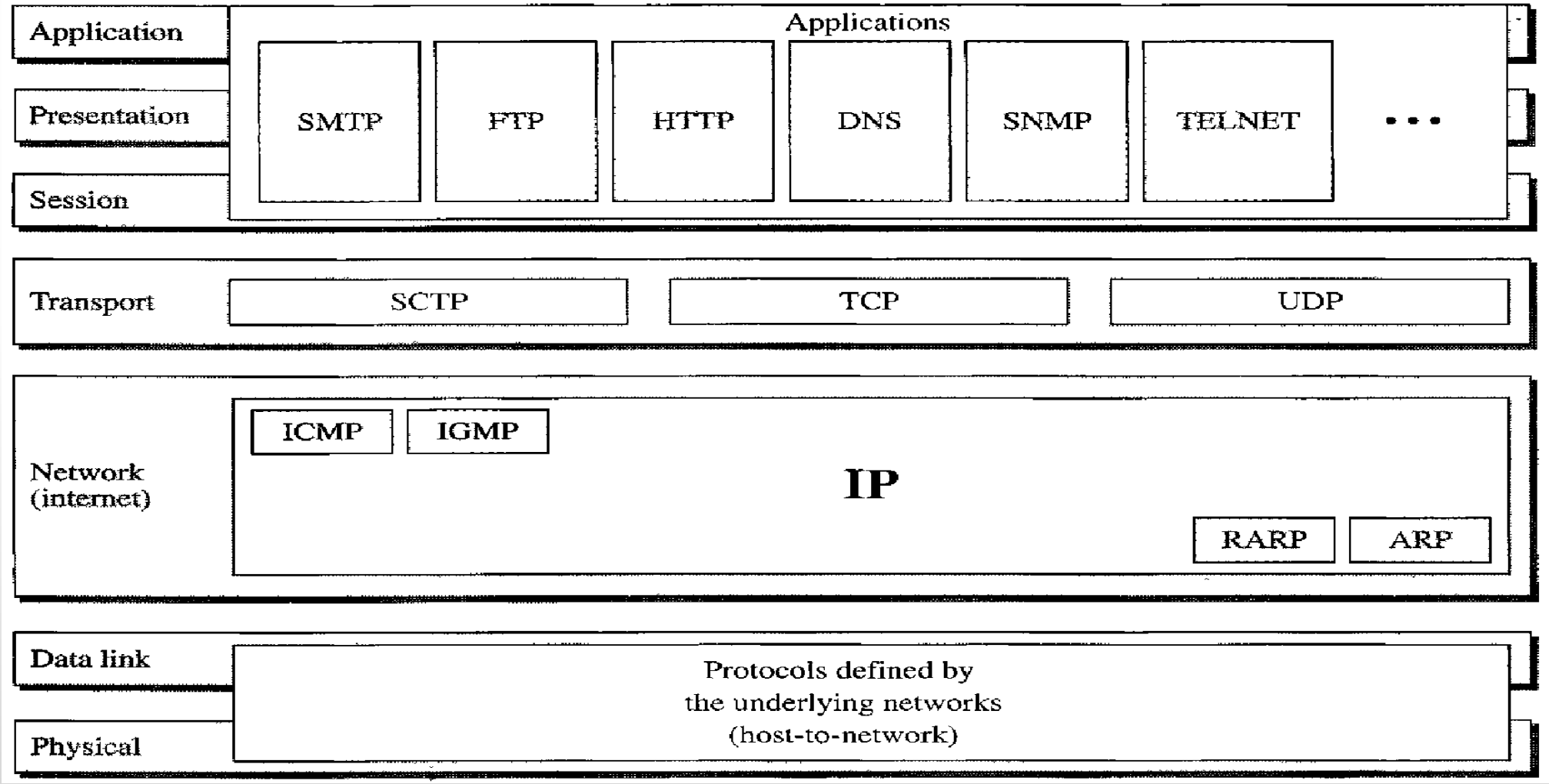
❖ Mapping of TCP/IP model and OSI model:





The TCP/IP Model: Mapping 2

❖ Mapping of TCP/IP model and OSI model:





The TCP/IP Model: Link Layer

Link Layer: Network Access Layer

- It does not define any specific protocol.
- Protocols define by underlying networks.
- Supports all the standard and proprietary protocols.
- Ethernet link Packets(Frame) sends over link.



The TCP/IP Model: Network Layer

Network Layer: Internet Layer

- Supports internetworking protocol.
- IP (Internetworking protocol) supports four protocols:
  1. ARP (Address Resolution Protocol)
  2. RARP (Reverse Address Resolution Protocol)
  3. ICMP (Internet Control Message Protocol)
  4. IGMP (Internet Group Message Protocol)



The TCP/IP Model: Network Layer

- **Network Layer:** Internet Layer
  - **Address Resolution Protocol**
    - ARP is used to associate a logical address with a physical address.
      - It will find the physical address of the node when its IP address is known.
  - **Reverse Address Resolution Protocol**
    - RARP allows a host to discover its **IP address** when it knows only its **physical address**.
      - ✓ It is used when a computer is connected to a network for the first time



The TCP/IP Model: Network Layer

Network Layer: Internet Layer

Internet Control Message Protocol

- ICMP is a mechanism use by the hosts and gateways to send notification of the datagram problems back to the sender.
  - ✓ ICMP sends query and error reporting messages.

Internet Group Message Protocol

- IGMP is used to facilitate the simultaneous transmission of a message to a group of recipients.



The TCP/IP Model: Transport Layer

- **Transport Layer:**
  - Initially, it has two protocols.
    1. User Datagram Protocol: UDP
    2. Transmission Control Protocol: TCP
  - New protocol called SCTP (Stream Control Transmission Protocol)



The TCP/IP Model: Transport Layer

- **Transport Layer:**
  - IP is a host-to-host protocol:
    - It can deliver a packet from one physical device to another.
  - UDP and TCP are transport level protocol
    - Responsible for delivery of a message from a process (running program) to another process.
  - A new transport layer protocol, SCTP has been devised to meet the needs of some newer applications.



The TCP/IP Model: Transport Layer

Transport Layer:

- It has two protocols.
  1. UDP: Process-to-process protocol that adds Port address, checksum, error control, & length information.
  2. TCP : Reliable stream transport protocol (stream means connection-oriented)

- New transport layer protocol called SCTP (Stream Control Transmission Protocol) devise to meet the needs of some newer applications such as VoIP etc.



The TCP/IP Model: Application Layer

- **Application Layer:** support services to users
  - Top layer of the Internet model.
  - Provide services used in everyday life like web client, email, etc.



Home Work

1. List down the similarities between OSI model and TCP/IP model.
2. List down dissimilarities between OSI model and TCP/IP model.



References

References:

1. Stallings, W., Data & Computer Communication, Prentice-Hall
2. Tanenbaum, Computers Networks, PHI
3. Forouzan, B.A., Data Communication & Networkings, 4th Edition.
4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan